

Foundation (Jalapeno)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q2.** Which of the following lines is perpendicular to $y = -3x - 2$?
(A) $y = \frac{1}{3}x + 1$
(B) $y = -\frac{1}{3}x - 1$
(C) $y = 3x - 2$
(D) $y = -3x + 2$
(E) $y = \frac{1}{3}x - 1$
- Q3.** If two lines have slopes m_1 and m_2 , what is the condition for them to be perpendicular?
(A) $m_1 = m_2$
(B) $m_1 = -m_2$
(C) $m_1 \cdot m_2 = -1$
(D) $m_1 \cdot m_2 = 1$
(E) $m_1 + m_2 = 0$
- Q4.** Find the equation of a line perpendicular to $y = x - 2$ passing through $(1, 3)$.
(A) $y = -x + 4$
(B) $y = x + 2$
(C) $y = -x - 2$
(D) $y = x - 4$
(E) $y = -x + 2$
- Q5.** If a line has a slope of -2 , what is the slope of a line perpendicular to it?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q6.** What is the equation of the line perpendicular to $y = 4x + 1$ and passing through $(-1, 2)$?
(A) $y = -\frac{1}{4}x + \frac{3}{2}$
(B) $y = -\frac{1}{4}x - \frac{3}{2}$
(C) $y = \frac{1}{4}x + \frac{3}{2}$
(D) $y = \frac{1}{4}x - \frac{3}{2}$
(E) $y = -4x + 1$
- Q7.** Two lines have equations $y = 2x + 1$ and $y = -\frac{1}{2}x + 3$. Are they perpendicular?
(A) Yes
(B) No
(C) Maybe
(D) Only if they intersect
(E) Only if they are parallel
- Q8.** Find the slope of the line perpendicular to $y = -\frac{3}{4}x + 2$.
(A) $\frac{4}{3}$
(B) $-\frac{4}{3}$
(C) $\frac{3}{4}$
(D) $-\frac{3}{4}$
(E) 0

Open Ended Questions (FRQ)

- Q9.** Prove that the lines $y = 2x + 1$ and $y = -\frac{1}{2}x - 3$ are not perpendicular. Show all work.
- Q10.** Find the equation of the line perpendicular to $y = x - 2$ and passing through $(2, 4)$. Show all work.